



FACULTY OF AGRICULTURE

BHABHA UNIVERSITY

Jatkhedi, NH-12 , Hoshangabad Road , Bhopal, Madhya Pradesh- 462026

Bachelor of Science in Agriculture B.Sc. Agriculture

SEMESTER-WISE DISTRIBUTION OF COURSES

SCHEME

SEMESTER-I			
S.NO.	SUBJECT TITLE	SUBJECT CODE	CREDITS
1.	Fundamentals of Horticulture	BAG-5101	2 (1+1)
2.	Fundamentals of Plant Biochemistry and Biotechnology	BAG-5102	3(2+1)
3.	Fundamentals of Soil Science	BAG-5103	3(2+1)
4.	Introduction to Forestry	BAG-5104	2 (1+1)
5.	Comprehension & Communication Skills in English	BAG-5105	2 (1+1)
6.	Fundamentals of Agronomy	BAG-5106	4(3+1)
7.	Introductory Biology*/Elementary Mathematics*	BAG-5107	2(1+1) /2(2+0)*
8.	Agricultural Heritage*	BAG-5108	1(1+0)*
9.	Rural Sociology & Educational Psychology	BAG-5109	2(2+0)
10.	Human Values & Ethics (non gradial)	BAG-5110	1(1+0)**
11.	NSS/NCC/Physical Education & Yoga Practices**	BAG-5111	2(0+2)**
TOTAL			18+03*+03**
*R: Remedial course; **NC: Non-gradial courses			

SEMESTER-II			
S.NO.	SUBJECT TITLE	SUBJECT CODE	CREDITS
1.	Fundamentals of Genetics	BAG-5201	3 (2+1)
2.	Agricultural Microbiology	BAG-5202	2(1+1)
3.	Soil and Water Conservation Engineering	BAG-5203	2(1+1)
4.	Fundamentals of Crop Physiology	BAG-5204	2 (1+1)
5.	Fundamentals of Agricultural Economics	BAG-5205	2 (2+0)
6.	Fundamentals of Plant Pathology	BAG-5206	4(3+1)
7.	Fundamentals of Entomology	BAG-5207	4(3+1)
8.	Fundamentals of Agricultural Extension Education	BAG-5208	3(2+1)
9.	Communication Skills and Personality Development	BAG-5209	2(1+1)
TOTAL			24(16+8)



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FIRST YEAR

SYLLABUS

SEMESTER I

SUBJECT TITLE	SUBJECT CODE	CREDIT
Fundamentals of Horticulture	BAG-5101	2 (1+1)

THEORY

UNIT I

Horticulture - Its definition and branches, importance and scope; horticultural and botanical classification; climate and soil for horticultural crops. Plant propagation-methods and propagating structures.

UNIT II

Seed dormancy, Seed germination, principles of orchard establishment; Principles and methods of training and pruning

UNIT III

Juvenility and flower bud differentiation; unfruitfulness; pollination, pollinizers and pollinators fertilization and parthenocarpy; medicinal and aromatic plants

UNIT IV

Importance of plant bio-regulators in horticulture

UNIT V

Irrigation – Methods, Fertilizer application in horticultural crops



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PRACTICAL

Identification of garden tools. Identification of horticultural crops. Preparation of seed bed/nursery bed. Practice of sexual and asexual methods of propagation including micro-propagation. Layout and planting of orchard. Training and pruning of fruit trees. Preparation of potting mixture. Fertilizer application in different crops. Visits to commercial nurseries/orchard.

FIRST YEAR

SYLLABUS

SEMESTER I

SUBJECT TITLE	SUBJECT CODE	CREDIT
Fundamentals of Plant Biochemistry and Biotechnology	BAG-5102	3 (2+1)

THEORY

Unit I

Importance of Biochemistry. Properties of Water, pH and Buffer. Carbohydrate: Importance and classification. Structures of Monosaccharides, Reducing and oxidizing properties of Monosaccharides, Mutarotation; Structure of Disaccharides and Poly saccharides. Lipid: Importance and classification; Structures and properties of fatty acids; storage lipids and membrane lipids.

Unit II

Proteins: Importance of proteins and classification; Structures, titration and zwitterions nature of amino acids; Structural organization of proteins. Enzymes: General properties; Classification; Mechanism of action; Michaelis&Menten and Line Weaver Burk equation & plots; Introduction to allosteric enzymes

Unit III

Nucleic acids: Importance and classification; Structure of Nucleotides, A, B & Z DNA; RNA: Types and Secondary & Tertiary structure. Metabolism of carbohydrates: Glycolysis, TCA cycle, Glyoxylate cycle, Electron transport chain. Metabolism of lipids: Beta oxidation, Biosynthesis of fatty acids.

Unit IV

Concepts and applications of plant biotechnology: Scope, organ culture, embryo culture, cell suspension culture, callus culture, anther culture, pollen culture and ovule culture and their applications; Micro-propagation methods; organogenesis and embryogenesis, Synthetic seeds and



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their significance; Embryo rescue and its significance; somatic hybridization and cybrids; Somaclonal variation and its use in crop improvement;

Unit V

Cryo-preservation; Introduction to recombinant DNA methods: physical (Gene gun method), chemical (PEG mediated) and Agrobacterium mediated gene transfer methods; Transgenics and its importance in crop improvement; PCR techniques and its applications; RFLP, RAPD, SSR; Marker Assisted Breeding in crop improvement; Biotechnology regulations.

PRACTICAL

Preparation of solution, pH & buffers, Qualitative tests of carbohydrates and amino acids. Quantitative estimation of glucose/ proteins. Titration methods for estimation of amino acids/lipids, Effect of pH, temperature and substrate concentration on enzyme action, Paper chromatography/ TLC demonstration for separation of amino acids/ Monosaccharides. Sterilization techniques. Composition of various tissue culture media and preparation of stock solutions for MS nutrient medium. Callus induction from various explants. Micro-propagation, hardening and acclimatization. Demonstration on isolation of DNA. Demonstration of gel electrophoresis techniques and DNA finger printing.

SYLLABUS

FIRST YEAR

SEMESTER I

SUBJECT TITLE	SUBJECT CODE	CREDIT
Fundamentals of Soil Science	BAG-5103	3(2+1)

THEORY

UNIT I

Soil- Pedological and edaphological concepts. Soil Science-scope and branches of soil science. Earth spheres and composition of earth crust. Minerals-classification, formation and properties of silicate and non silicate minerals, Rocks-classification, formation and properties of igneous, sedimentary and metamorphic rocks.

UNIT II

Weathering-type, factors of weathering, products of weathering; Soil formation-soil forming factors and soil forming processes. Soil profile-master horizons, subordinate horizons. Soil physical properties-Soil texture-classification of soil separates, properties of soil separates, Particle size analysis- Stokes law-assumptions and limitations, textural classes.



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UNIT III

Soil structure-classification, soil aggregates, evaluation of soil structure, significance. Pore space- types, factors affecting porosity, manipulation. Bulk density and particle density-relationships, factors, significance and manipulation. Soil colour-factors, attributes and significance. Soil consistency- forms, factors, limits and significance.

UNIT IV

Soil crusting- factors and significance. Soil temperature-thermal properties of soils, flow of heat, soil temperature regimes, influence of soil temperature on plant growth. Soil air-composition, gaseous exchange, influence of soil air on plant growth.

UNIT V

Soil water-classification, potentials, soil moisture constants, movement of soil water, infiltration, percolation, hydraulic conductivity. Soil survey- types and methods. Soil classification-systems of classification. Soil taxonomy-advantages, structure, formative elements, diagnostic horizons, keys to soil orders. Soils of Karnataka and India.

PRACTICAL

- Study of general properties of minerals
- Study of minerals-silicate and non-silicate minerals
- Study of rocks-igneous, sedimentary and metamorphic rocks
- Study of a soil profile
- Collection and processing of soil for analysis
- Study of soil texture-feel method, mechanical analysis
- Determination of bulk density, particle density and soil porosity
- Determination of soil colour
- Study of soil structure and aggregate analysis
- Determination of soil moisture
- Determination of soil moisture constants-field capacity; water holding capacity
- Study of infiltration rate of soil
- Study of soil temperature



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FIRST YEAR

SYLLABUS

SEMESTER I

SUBJECT TITLE	SUBJECT CODE	CREDIT
Introduction to Forestry	BAG-5104	2(1+1)

THEORY

UNIT I

Introduction – definitions of basic terms related to forestry, objectives of silviculture, forest classification, salient features of Indian Forest Policies.

UNIT II

Agroforestry – definitions, importance, criteria of selection of trees in agroforestry, different agroforestry systems prevalent in the country, shifting cultivation, taungya, alley cropping, wind breaks and shelter belts, home gardens. Cultivation practices of two important fast growing tree species of the region.

UNIT III

Forest regeneration, Natural regeneration - natural regeneration from seed and vegetative parts, coppicing, pollarding, root suckers; Artificial regeneration – objectives, choice between natural and artificial regeneration, essential preliminary considerations.

UNIT IV

Crown classification. Tending operations – weeding, cleaning, thinning – mechanical, ordinary, crown and advance thinning.

UNIT V

Forest mensuration – objectives, diameter measurement, instruments used in diameter measurement; Non instrumental methods of height measurement - shadow and single pole method; Instrumental methods of height measurement - geometric and trigonometric principles, instruments used in height measurement; tree stem form, form factor, form quotient, measurement of volume of felled and standing trees, age determination of trees.

PRACTICAL

- Identification of tree-species
- Diameter measurements using calipers and tape



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- Diameter measurements of forked, buttressed, fluted and leaning trees
- Height measurement of standing trees by shadow method, single pole method and hypsometer
- Volume measurement of logs using various formulae
- Nursery lay out, seed sowing, vegetative propagation techniques
- Forest plantations and their management
- Visits of nearby forest based industries

SYLLABUS

FIRST YEAR

SEMESTER I

SUBJECT TITLE	SUBJECT CODE	CREDIT
Comprehension & Communication Skills in English	BAG-5105	2(1+1)

THEORY

Unit I

War Minus Shooting- The sporting Spirit. A Dilemma- A layman looks at science Raymond B. Fosdick. You and Your English Spoken English and broken English G.B. Shaw.

Unit II

Reading Comprehension, Vocabulary- Antonym, Synonym, Homophones, Homonyms, often confused words. Exercises to Help the students in the enrichment of vocabulary based on TOEFL and other competitive examinations.

Unit III

Functional grammar: Articles, Prepositions, Verb, Subject verb Agreement, Transformation, Synthesis, Direct and Indirect Narration.

Unit IV

Written Skills: Paragraph writing, Precise writing, Report writing and Proposal writing. The Style: Importance of professional writing.

Unit V

Preparation of Curriculum Vitae and Job applications. Synopsis Writing. Interviews: kinds, Importance and process.



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PRACTICAL

Listening Comprehension: Listening to short talks lectures, speeches (scientific, commercial and general in nature). Oral Communication: Phonetics, stress and intonation, Conversation practice. Conversation: rate of speech, clarity of voice, speaking and Listening, politeness & Reading skills: reading dialogues, rapid reading, intensive reading, improving reading skills. Mock Interviews: testing initiative, team spirit, leadership, intellectual ability. Group Discussions.

FIRST YEAR

SYLLABUS

SEMESTER I

SUBJECT TITLE	SUBJECT CODE	CREDIT
Fundamentals of Agronomy	BAG-5106	4(3+1)

THEORY

Unit I

Agronomy and its scope, seeds and sowing, tillage and tilth, crop density and geometry.

Unit II

Crop nutrition, manures and fertilizers, nutrient use efficiency.

Unit III

water resources, soil-plant-water relationship, crop water requirement, water use efficiency, irrigation- scheduling criteria and methods, quality of irrigation water, logging.

Unit IV

Weeds- importance, classification, crop weed competition, concepts of weed management principles and methods, herbicides- classification, selectivity and resistance, allelopathy.

Unit V

Growth and development of crops, factors affecting growth and development, plant ideotypes, crop rotation and its principles, adaptation and distribution of crops, crop management technologies in problematic areas, harvesting and threshing of crops.

PRACTICAL

Identification of crops, seeds, fertilizers, pesticides and tillage implements, study of agroclimatic



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zones of India, Identification of weeds in crops, Methods of herbicide and fertilizer application, Study of yield contributing characters and yield estimation, Seed germination and viability test, Numerical exercises on fertilizer requirement, plant population, herbicides and water requirement, Use of tillage implements-reversible plough, one way plough, harrow, leveler, seed drill, Study of soil moisture measuring devices, Measurement of field capacity, bulk density and infiltration rate, Measurement of irrigation water.

FIRST YEAR

SYLLABUS

SEMESTER I

SUBJECT TITLE	SUBJECT CODE	CREDIT
Introductory Biology*/Elementary Mathematics*	BAG-5107	2(1+1) /2(2+0)*

INTRODUCTORY BIOLOGY (NEW) 2(1+1)

THEORY

Unit I

Introduction to the living world, diversity and characteristics of life, origin of life.

Unit II

Evolution and Eugenics.

Unit III

Binomial nomenclature and classification Cell and cell division.

Unit IV

Morphology of flowering plants. Seed and seed germination. Plant systematic- viz; Brassicaceae, Fabaceae and Poaceae.

Unit V

Role of animals in agriculture.

PRACTICAL

Morphology of flowering plants – root, stem and leaf and their modifications. Inflorescence, flower and fruits. Cell, tissues & cell division. Internal structure of root, stem and leaf. Study of specimens and slides. Description of plants - Brassicaceae, Fabaceae and Poaceae.



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ELEMENTARY MATHEMATICS (NEW) 2(2+0)

THEORY

Unit I

Straight lines : Distance formula, section formula (internal and external division), Change of axes (only origin changed), Equation of co-ordinate axes, Equation of lines parallel to axes, Slope- intercept form of equation of line, Slope-point form of equation of line, Two point form of equation of line, Intercept form of equation of line, Normal form of equation of line, General form of equation of line, Point of intersection of two st. lines, Angles between two st. lines, Parallel lines, Perpendicular lines, Angle of bisectors between two lines, Area of triangle and quadrilateral.

Unit II

Circle: Equation of circle whose centre and radius is known, General equation of a circle, Equation of circle passing through three given points, Equation of circle whose diameter is line joining two points (x_1, y_1) & (x_2, y_2) , Tangent and Normal to a given circle at given point (Simple problems), Condition of tangency of a line $y = mx + c$ to the given circle $x^2 + y^2 = a^2$.

Unit III

Differential Calculus: Definition of function, limit and continuity, Simple problems on limit, Simple problems on continuity, Differentiation of x^n , e^x , $\sin x$ & $\cos x$ from first principle, Derivatives of sum, difference, product and quotient of two functions, Differentiation of functions of functions (Simple problem based on it), Logarithmic differentiation (Simple problem based on it), Differentiation by substitution method and simple problems based on it, Differentiation of Inverse Trigonometric functions. Maxima and Minima of the functions of the form $y=f(x)$ (Simple problems based on it).

Unit IV

Integral Calculus : Integration of simple functions, Integration of Product of two functions, Integration by substitution method, Definite Integral (simple problems based on it), Area under simple well-known curves (simple problems based on it).

Unit V

Matrices and Determinants: Definition of Matrices, Addition, Subtraction, Multiplication, Transpose and Inverse up to 3rd order, Properties of determinants up to 3rd order and their evaluation.



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FIRST YEAR

SYLLABUS

SEMESTER I

SUBJECT TITLE	SUBJECT CODE	CREDIT
Agricultural Heritage (New Course)*	BAG-5108	1(1+0)*

THEORY

Unit I

Introduction of Indian agricultural heritage; Ancient agricultural practices, Relevance of heritage to present day agriculture; Past and present status of agriculture and farmers in society;

Unit II

Journey of Indian agriculture and its development from past to modern era; Plant production and protection through indigenous traditional knowledge;

Unit III

Crop voyage in India and world; Agriculture scope; Importance of agriculture and agricultural resources available in India;

Unit IV

Crops significance and classifications; National agriculture setup in India;

Unit V

Current scenario of Indian agriculture; Indian agricultural concerns and future prospects.

FIRST YEAR

SYLLABUS

SEMESTER I

SUBJECT TITLE	SUBJECT CODE	CREDIT
Rural Sociology & Educational Psychology	BAG-5109	2(2+0)*



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THEORY

UNIT I

Sociology and Rural sociology: Definition and scope, its significance in agriculture extension, Social Ecology.

UNIT II

Rural society, Social Groups, Social Stratification, Culture concept, Social Institution, Social Change & Development.

UNIT III

Educational psychology: Meaning & its importance in agriculture extension.

UNIT IV

Behavior: Cognitive, affective, psychomotor domain, Personality, Learning,

UNIT V

Motivation, Theories of Motivation, Intelligence.

SYLLABUS

FIRST YEAR

SEMESTER I

SUBJECT TITLE	SUBJECT CODE	CREDIT
Human Values & Ethics (non gradial)	BAG-5110	1(0+1)

THEORY

Unit I

Values and Ethics-An Introduction.

Unit II

Goal and Mission of Life.Vision of Life.Principles and Philosophy.

Unit III

Self-Exploration,Self-Awareness,Self-Satisfaction, Decision Making.

Unit IV

Motivation, Sensitivity, Success, Selfless Service, Case Study of Ethical Lives, Positive Spirit.



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Unit V

Body, Mind and Soul, Attachment and Detachment. Spirituality Quotient, Examination.

SYLLABUS

FIRST YEAR

SEMESTER I

SUBJECT TITLE	SUBJECT CODE	CREDIT
NSS/NCC/Physical Education & Yoga Practices**	BAG-5111	2(0+2)

SEMESTER I

COURSE TITLE: NATIONAL SERVICE SCHEME I

Introduction and basic components of NSS:

Orientation: history, objectives, principles, symbol, badge; programs
NSS, 113 organizational structure of NSS, code of conduct for NSS volunteers, points to be
considered by NSS volunteers awareness about health r

NSS programs and activities

Concept of regular activities, special camping, day camps, basis of adoption of village/slums, conducting survey, analyzing guiding financial patterns of scheme, youth programme/ schemes of GOI, coordination with different agencies and maintenance of diary

Understanding youth

Definition, profile, categories, issues and challenges of youth; and opportunities for youth who is agent of the social change

Community mobilization

Mapping of community stakeholders, designing the message as per problems and their culture; identifying methods of mobilization involving youth-adult partnership

Social harmony and national integration

Indian history and culture, role of youth in nation building, conflict resolution and peacebuilding

Volunteerism and shramdan

Indian tradition of volunteerism, its need, importance, motivation and constraints; shramdan as part of volunteerism

Citizenship, constitution and human rights



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Basic features of constitution of India, fundamental rights and duties, human rights, consumer awareness and rights and rights to information

Family and society

Concept of family, community (PRIs and other community based organizations) and society



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SYLLABUS

FIRST YEAR

SEMESTER II

SUBJECT TITLE	SUBJECT CODE	CREDIT
Fundamentals of Genetics	BAG-5201	3 (2+1)

THEORY

Unit I

Pre and Post Mendelian concepts of heredity, Mendelian principles of heredity.

Unit II

Architecture of chromosome; chromonemata, chromosome matrix, chromomeres, centromere, secondary constriction and telomere; special types of chromosomes. Chromosomal theory of inheritance, cell cycle and cell division- mitosis and meiosis. Probability and Chi-square. Dominance relationships, Epistatic interactions with example.

Unit III

Multiple alleles, pleiotropism and pseudoalleles, Sex determination and sex linkage, sex limited and sex influenced traits, Blood group genetics, Linkage and its estimation, crossing over mechanisms, chromosome mapping. Structural and numerical variations in chromosome and their implications,

Unit IV

Use of haploids, dihaploids and doubled haploids in Genetics, Mutation, classification, Methods of inducing mutations & CIB technique, mutagenic agents and induction of mutation, Qualitative & Quantitative traits,

Unit V

Polygenes and continuous variations, multiple factor hypothesis, Cytoplasmic inheritance, Genetic disorders, Nature, structure & replication of genetic material. Protein synthesis, Transcription and translational mechanism of genetic material, Gene concept: Gene structure, function and regulation, Lac and Trp operons.

PRACTICAL

Study of microscope, Study of cell structure. Mitosis and Meiosis cell division. Experiments on monohybrid, dihybrid, trihybrid, test cross and back cross, Experiments on epistatic interactions



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including test cross and back cross, Practice on mitotic and meiotic cell division, Experiments on probability and Chi-square test. Determination of linkage and cross-over analysis (through two point test cross and three point test cross data). Study on sex linked inheritance in *Drosophila*. Study of models on DNA and RNA structures.



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FIRST YEAR SYLLABUS SEMESTER II

SUBJECT TITLE	SUBJECT CODE	CREDIT
Agricultural Microbiology	BAG-5202	2(1+1)

THEORY

Unit 1

Introduction, Microbial world: Prokaryotic and eukaryotic microbes. Bacteria: cell structure, chemoautotrophy, photo autotrophy, growth

Unit 2

Bacterial genetics: Genetic recombination transformation, conjugation and transduction, plasmids, transposon.

Unit 3

Role of microbes in soil fertility and crop production:

Unit 4

Carbon, Nitrogen, Phosphorus and Sulphur cycles. Biological nitrogen fixation- symbiotic, associative and asymbiotic.

Unit 5

Azolla, blue green algae and mycorrhiza. Rhizosphere and phyllosphere. Microbes in human welfare: silage production, biofertilizers, biopesticides, biofuel production and biodegradation of agro-waste.

PRACTICAL

Introduction to microbiology laboratory and its equipments; Microscope- parts, principles of microscopy, resolving power and numerical aperture. Methods of sterilization. Nutritional media and their preparations. Enumeration of microbial population in soil- bacteria, fungi, actinomycetes. Methods of isolation and purification of microbial cultures. Isolation of *Rhizobium* from legume root nodule. Isolation of *Azotobacter* from soil. Isolation of *Azospirillum* from roots. Isolation of BGA. Staining and microscopic examination of microbes.



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FIRST YEAR

SYLLABUS

SEMESTER II

SUBJECT TITLE	SUBJECT CODE	CREDIT
Soil and Water Conservation Engineering	BAG-5203	2(1+1)

THEORY

Unit I

Introduction to Soil and Water Conservation, causes of soil erosion. Definition and agents of soil erosion, water erosion: Forms of water erosion.

Unit II

Gully classification and control measures. Soil loss estimation by universal Loss Soil Equation. Soil loss measurement techniques.

Unit III

Principles of erosion control: Introduction to contouring, strip cropping. Contour bund. Graded bund and bench terracing.

Unit IV

Grassed water ways and their design. Water harvesting and its techniques.

Unit V

Wind erosion: mechanics of wind erosion, types of soil movement. Principles of wind erosion control and its control measures.

PRACTICAL

General status of soil conservation in India, Calculation of erosion index, Estimation of soil loss, Measurement of soil loss, Preparation of contour maps. Design of grassed water ways, Design of contour bunds, Design of graded bunds, Design of bench terracing system, Problem on wind erosion.



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SEMESTER II

SUBJECT TITLE	SUBJECT CODE	CREDIT
Fundamentals of Crop Physiology	BAG-5204	2(1+1)

THEORY

Unit I

Introduction to crop physiology and its importance in Agriculture;

Unit II

Plant cell: an Overview; Diffusion and osmosis; Absorption of water, transpiration and Stomatal Physiology; Mineral nutrition of Plants: Functions and deficiency symptoms of nutrients, nutrient uptake mechanisms; Photosynthesis:

Unit III

Light and Dark reactions, C₃, C₄ and CAM plants; Respiration: Glycolysis, TCA cycle and electron transport chain; Fat Metabolism:

Unit IV

Fatty acid synthesis and Breakdown; Plant growth regulators:

Unit V

Physiological roles and agricultural uses, Physiological aspects of growth and development of major crops: Growth analysis, Role of Physiological growth parameters in crop productivity.

PRACTICAL

Study of plant cells, structure and distribution of stomata, imbibitions, osmosis, plasmolysis, measurement of root pressure, rate of transpiration, Separation of photosynthetic pigments through paper chromatography, Rate of transpiration, photosynthesis, respiration, tissue test for mineral nutrients, estimation of relative water content, Measurement of photosynthetic CO₂ assimilation by Infra Red Gas Analyser (IRGA).



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SEMESTER II

SUBJECT TITLE	SUBJECT CODE	CREDIT
Fundamentals of Agricultural Economics	BAG-5205	2 (2+0)

THEORY

UNIT I

Economics: Meaning, scope and subject matter, definitions, activities, approaches to economic analysis; micro and macro economics, positive and normative analysis. Nature of economic theory; rationality assumption, concept of equilibrium, economic laws as generalization of human behavior. Basic concepts: Goods and services, desire, want, demand, utility, cost and price, wealth, capital, income and welfare.

UNIT II

Agricultural economics: meaning, definition, characteristics of agriculture, importance and its role in economic development. Agricultural planning and development in the country. *Demand:* meaning, law of demand, schedule and demand curve, determinants, utility theory; law of diminishing marginal utility, equi-marginal utility principle. Consumer's equilibrium and derivation of demand curve, concept of consumer surplus. Elasticity of demand: concept and measurement of price elasticity, income elasticity and cross elasticity.

UNIT III

Production: process, creation of utility, factors of production, input output relationship. *Laws of returns:* Law of variable proportions and law of returns to scale. Cost: concepts, short run and long run cost curves. Supply: Stock v/s supply, law of supply, schedule, supply curve, determinants of supply, elasticity of supply.

UNIT IV

Market structure: meaning and types of market, basic features of perfectly competitive and imperfect markets. Price determination under perfect competition; short run and long run equilibrium of firm and industry, shut down and break even points. Distribution theory: meaning, factor market and pricing of factors of production. Concepts of rent, wage, interest and profit.

UNIT V

National income: Meaning and importance, circular flow, concepts of national income accounting and approaches to measurement, difficulties in measurement. Population: Importance, Malthusian



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and Optimum population theories, natural and socio economic determinants, current policies and programmes on population control. Money: Barter system of exchange and its problems, evolution, meaning and functions of money, classification of money, supply, general price index, inflation and deflation. Banking: Role in modern economy, types of banks, functions of commercial and central bank, credit creation policy. Agricultural and public finance: meaning, micro v/s macro finance, need for agricultural finance, public revenue and public expenditure. Tax: meaning, direct and indirect taxes, agricultural taxation, VAT. Economic systems: Concepts of economy and its functions, important features of capitalistic, socialistic and mixed economies, elements of economic planning.

FIRST YEAR

SYLLABUS

SEMESTER II

SUBJECT TITLE	SUBJECT CODE	CREDIT
Fundamentals of Plant Pathology	BAG-5206	4 (3+1)

THEORY

UNIT I

Introduction: Importance of plant diseases, scope and objectives of Plant Pathology. History of Plant Pathology with special reference to Indian work. Terms and concepts in Plant Pathology. Pathogenesis. Causes / factors affecting disease development: disease triangle and tetrahedron and classification of plant diseases. Important plant pathogenic organisms, different groups: fungi, bacteria, fastidious vesicular bacteria, phytoplasmas, spiroplasmas, viruses, viroids, algae, protozoa, phanerogamic parasites and nematodes with examples of diseases caused by them. Diseases and symptoms due to abiotic causes.

UNIT II

Fungi: general characters, definition of fungus, somatic structures, types of fungal thalli, fungal tissues, modifications of thallus, reproduction (asexual and sexual). Nomenclature, Binomial system of nomenclature, rules of nomenclature, classification of fungi. Key to divisions, subdivisions, orders and classes.

UNIT III

Bacteria and mollicutes: general morphological characters. Basic methods of classification and reproduction. *Viruses:* nature, structure, replication and transmission. Study of phanerogamic plant parasites.



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UNIT IV

Nematodes: General morphology and reproduction, classification, symptoms and nature of damage caused by plant nematodes (*Heterodera*, *Meloidogyne*, *Anguina*, *Radopholus* etc.)

UNIT V

Growth and reproduction of plant pathogens. Liberation / dispersal and survival of plant pathogens. Types of parasitism and variability in plant pathogens. Pathogenesis. Role of enzymes, toxins and growth regulators in disease development. Defense mechanism in plants. Epidemiology: Factors affecting disease development. Principles and methods of plant disease management. Nature, chemical combination, classification, mode of action and formulations of fungicides and antibiotics.

PRACTICAL

Acquaintance with various laboratory equipments and microscopy. Collection and preservation of disease specimen. Preparation of media, isolation and Koch's postulates. General study of different structures of fungi. Study of symptoms of various plant diseases. Study of representative fungal genera. Staining and identification of plant pathogenic bacteria. Transmission of plant viruses. Study of phanerogamic plant parasites.

Study of morphological features and identification of plant parasitic nematodes. Sampling and extraction of nematodes from soil and plant material, preparation of nematode mounting.

Study of fungicides and their formulations. Methods of pesticide application and their safe use. Calculation of fungicide sprays concentrations.

SYLLABUS

FIRST YEAR

SEMESTER II

SUBJECT TITLE	SUBJECT CODE	CREDIT
Fundamentals of Entomology	BAG-5207	4 (3+1)

THEORY

UNIT I

History of Entomology in India. Major points related to dominance of Insecta in Animal kingdom. Classification of phylum Arthropoda upto classes. Relationship of class Insecta with other classes of Arthropoda. Morphology: Structure and functions of insect cuticle and molting.



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UNIT II

Body segmentation, Structure of Head, thorax and abdomen. Structure and modifications of insect antennae, mouth parts, legs, Wing venation, modifications and wing coupling apparatus. Structure of male and female genital organ.

UNIT III

Metamorphosis and diapause in insects. Types of larvae and pupae. Structure and functions of digestive, circulatory, excretory, respiratory, nervous, secretory (Endocrine) and reproductive system, in insects. Types of reproduction in insects. Major sensory organs like simple and compound eyes, chemoreceptor. Insect Ecology: Introduction, Environment and its components. Effect of abiotic factors— temperature, moisture, humidity, rainfall, light, atmospheric pressure and air currents. Effect of biotic factors – food competition, natural and environmental resistance.

UNIT IV

Categories of pests. Concept of IPM, Practices, scope and limitations of IPM. Classification of insecticides, toxicity of insecticides and formulations of insecticides. Chemical control, importance, hazards and limitations. Recent methods of pest control, repellents, anti feed ants, hormones, attractants, gamma radiation. Insecticides Act 1968- Important provisions. Application techniques of spray fluids. Symptoms of poisoning, first aid and antidotes.

UNIT V

Systematics: Taxonomy –importance, history and development and binomial nomenclature. Definitions of Biotype, Sub-species, Species, Genus, Family and Order. Classification of class Insecta upto Orders, basic groups of present day insects with special emphasis to orders and families of Agricultural importance like Orthoptera: Acrididae, Tettigonidae, Gryllidae, Gryllotalpidae; Dictyoptera: Mantidae, Blattidae; Odonata; Isoptera: Termitidae; Thysanoptera: Thripidae; Hemiptera: Pentatomidae, Coreidae, Cimicidae, Pyrrhocoridae, Lygaeidae, Cicadellidae, Delphacidae, Aphididae, Coccidae, Lophophidae, Aleurodidae, Pseudococcidae; Neuroptera: Chrysopidae; Lepidoptera: Pieridae, Papilionidae, Noctuidae, Sphingidae, Pyralidae, Gelechiidae, Arctiidae, Saturnidae, Bombycidae; Coleoptera: Coccinellidae, Chrysomelidae, Cerambycidae, Curculionidae, Bruchidae, Scarabaeidae; Hymenoptera: Tenthredinidae, Apidae. Trichogrammatidae, Ichneumonidae, Braconidae, Chalcididae; Diptera: Cecidomyiidae, Tachinidae, Agromyziidae, Culicidae, Muscidae, Tephritidae.

PRACTICAL

Methods of collection and preservation of insects including immature stages; External features of Grasshopper/Blister beetle; Types of insect antennae, mouthparts and legs; Wing venation, types of wings and wing coupling apparatus. Types of insect larvae and pupae; Dissection of digestive system in insects (Grasshopper); Dissection of male and female reproductive systems in insects



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(Grasshopper); Study of characters of orders Orthoptera, Dictyoptera, Odonata, Isoptera, Thysanoptera, Hemiptera, Lepidoptera, Neuroptera, Coleoptera, Hymenoptera, Diptera and their families of agricultural importance. Insecticides and their formulations. Pesticide appliances and their maintenance. Sampling techniques for estimation of insect population and damage.

SYLLABUS

FIRST YEAR

SEMESTER II

SUBJECT TITLE	SUBJECT CODE	CREDIT
Fundamentals of Agricultural Extension Education	BAG-5208	3(2+1)

THEORY

Unit I

Education: Meaning, definition & Types; Extension Education- meaning, definition, scope and process; objectives and principles of Extension Education; Extension Programme planning- Meaning, Process, Principles and Steps in Programme Development.

Unit II

Extension systems in India: extension efforts in pre-independence era (Sriniketan, Marthandam, Firka Development Scheme, Gurgaon Experiment, etc.) and post-independence era (Etawah Pilot Project, Nilokheri Experiment, etc.); various extension/ agriculture development programmes launched by ICAR/ Govt. of India (IADP, IAAP, HYVP, KVK, IVLP, ORP, ND, NATP, NAIP, etc.).

Unit III

New trends in agriculture extension: privatization extension, cyber extension/ e-extension, market-led extension, farmer-led extension, expert systems, etc. Rural Development: concept, meaning, definition; various rural development programmes launched by Govt. of India. Community Dev.- meaning, definition, concept & principles, Philosophy of C.D.

Unit IV

Rural Leadership: concept and definition, types of leaders in rural context; extension administration: meaning and concept, principles and functions. Monitoring and evaluation: concept and definition, monitoring and evaluation of extension programmes; transfer of technology: concept and models, capacity building of extension personnel; extension teaching methods: meaning, classification, individual, group and mass contact methods, ICT Applications in TOT (New and Social Media), media mix strategies;

Unit V



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communication: meaning and definition; Principles and Functions of Communication, models and barriers to communication. Agriculture journalism; diffusion and adoption of innovation: concept and meaning, process and stages of adoption, adopter categories.

PRACTICAL

To get acquainted with university extension system. Group discussion- exercise; handling and use of audio visual equipments and digital camera and LCD projector; preparation and use of AV aids, preparation of extension literature – leaflet, booklet, folder, pamphlet news stories and success stories; Presentation skills exercise; micro teaching exercise; A visit to village to understand the problems being encountered by the villagers/ farmers; to study organization and functioning of DRDA and other development departments at district level; visit to NGO and learning from their experience in rural development; understanding PRA techniques and their application in village development planning; exposure to mass media: visit to community radio and television studio for understanding the process of programme production; script writing, writing for print and electronic media, developing script for radio and television.

SYLLABUS

FIRST YEAR

SEMESTER II

SUBJECT TITLE	SUBJECT CODE	CREDIT
Communication Skills and Personality Development	BAG-5209	2(1+1)

THEORY

Unit I Communication Skills: Structural and functional grammar

Unit II Meaning and process of communication, verbal and nonverbal communication

Unit III Listening and note taking, writing skills, oral presentation skills; field diary and lab record; indexing, footnote and bibliographic procedures.

Unit IV Reading and comprehension of general and technical articles, precise writing, summarizing, abstracting

Unit V Individual and group presentations, impromptu presentation, public speaking; Group discussion, Organizing seminars and conferences.



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PRACTICAL

Listening and note taking, writing skills, oral presentation skills; field diary and lab record; indexing, footnote and bibliographic procedures. Reading and comprehension of general and technical articles, precise writing, summarizing, abstracting; individual and group presentations.